



The Z-Score Journey

Section 1: Learn

What is Z-Score?

Z-score, also known as the **standard score**, measures **how many standard deviations** a data point is from the **mean**.

Why is Z-Score Important?

- **Standardizes Data** – Makes different datasets comparable.
- **Identifies Outliers** – Helps detect unusual data points.
- **Used in Probability & Statistics** – Helps calculate **percentiles** and **probabilities**.

Z-Score Formula

$$Z = \frac{x - \mu}{\sigma}$$

Diagram illustrating the Z-score formula with labels and arrows:

- Score** points to x .
- Mean** points to μ .
- SD** (Standard Deviation) points to σ .

- **z**: The Z-score (standard score).
- **x**: The individual data point or raw score you're evaluating.
- **μ (mu)**: The mean (average) of the entire dataset.
- **σ (sigma)**: The standard deviation of the dataset, measuring data spread.

Interpreting Z-Score

- **Z = 0** → Data point is **equal to the mean**.



- $Z > 0 \rightarrow$ Data point is **above the mean**.
- $Z < 0 \rightarrow$ Data point is **below the mean**.
- $Z > 2$ or $Z < -2 \rightarrow$ Potential **outlier**.



How is Z-Score Used?

- **Student Performance** – Comparing exam scores from different subjects.
- **Stock Market** – Identifying overbought or oversold stocks.
- **Health Sector** – Diagnosing abnormal cholesterol levels.

Anecdote: How Z-Score Helped in College Admissions

A university was receiving **applications from students worldwide**.

To compare students **fairly**, they converted all marks into **Z-scores**.

This ensured students from **different education boards** were **evaluated equally**.

Section 2: Practice

fi. Calculating Z-Score for an Exam Score

Example: Class Test Scores

Stud ent	Mar ks
A	55
B	60
C	70
D	85
E	90

Given:

- **Mean (μ) = 72**
- **Standard Deviation (σ) = 10**



For **Student B (Score = 60)**:

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$$Z = \frac{60 - 72}{10} = \frac{-12}{10} = -1.2$$

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Interpretation:

- A Z-score of **-1.2** means Student B's score is **1.2 standard deviations below the class average**.
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2. Finding Outliers in a Dataset

Example: House Prices in a Locality (₹ in Lakhs)

House	Price
A	50
B	55
C	53
D	48
E	150

- Mean (μ) = **71.2**
- Standard Deviation (σ) = **41.5**
- **Z-score of House E =**

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$$Z = \frac{150 - 71.2}{41.5} = 1.89$$

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Since **$Z > 2$** , House E is an **outlier**.



Section 3: Know More

Frequently Asked Questions (FAQs)

fi. What does a Z-Score of 2 mean?

A Z-score of 2 means the data point is **2 standard deviations above the mean**.

2. Can Z-Score be negative?

Yes, a **negative Z-score** means the value is **below the mean**.

3. What is a good Z-Score?

A Z-score between **-2 and 2** is usually considered **normal**.

4. How does Z-Score help in standardization?

Z-score **converts different datasets into a common scale**, making them **comparable**.

5. When should I use Z-Score?

- **Comparing different distributions** (e.g., different exam grading systems).
- **Detecting outliers** (e.g., abnormal medical test results).

Conclusion:

Z-score is a **powerful statistical tool** that helps in **standardization, comparison, and anomaly detection**.

By converting values into a **common scale**, Z-score **simplifies data interpretation and decision-making**.